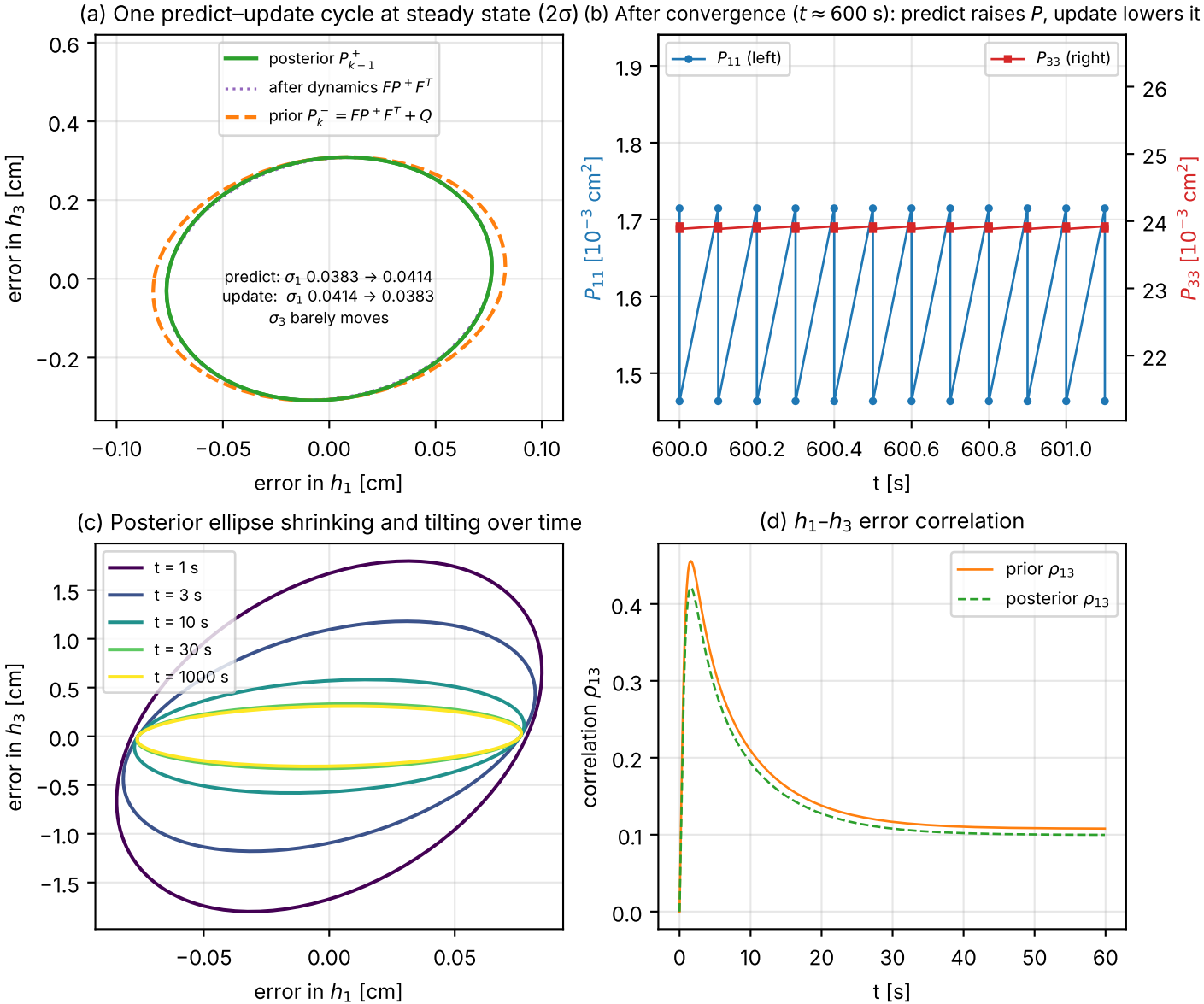
(a) One predict–update cycle at steady state (2σ)
posterior P_{k-1}^{+}
after dynamics $FP^{+}F^{T}$
prior $P_k^{-} = FP^{+}F^{T} + Q$
predict: σ_1 0.0383 → 0.0414
update: σ_1 0.0414 → 0.0383
σ_3 barely moves
error in h_3 [cm]
error in h_1 [cm]
(b) After convergence ($t \approx 600$ s): predict raises P, update lowers it
P_{11} (left)
P_{33} (right)
P_{11} [10^{-3} cm^2]
P_{33} [10^{-3} cm^2]
t [s]
(c) Posterior ellipse shrinking and tilting over time
t = 1 s
t = 3 s
t = 10 s
t = 30 s
t = 1000 s
error in h_3 [cm]
error in h_1 [cm]
(d) h_1–h_3 error correlation
prior ρ_{13}
posterior ρ_{13}
correlation ρ_{13}
t [s]